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A hybrid system with optimized decomposition on random deep learning model for crude oil futures forecasting

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ABSTRACT

The violent fluctuations in crude oil prices have a significant influence on the global economy. Therefore, it is crucial to establish a precise and reliable model for forecasting crude oil prices. To enhance the forecasting accuracy of crude oil futures prices, this study introduces an innovative hybrid approach. The proposed approach integrates optimized variational mode decomposition, an attention mechanism and the stochastic error correction algorithm into the deep bidirectional long short term memory network. First, a novel decomposition approach is introduced that incorporates the concept of cross fuzzy entropy into variational mode decomposition. Second, an attention mechanism is employed to improve the effectiveness of the forecasting framework. Third, the stochastic strength formula is employed to correct errors during the model training phase, consequently strengthening the acquisition of efficient characteristics through historical data weight adjustment. Finally, two sets of crude oil futures sequences are applied for empirical predictions. The proposed model exhibits outstanding performance by studying the results of multiple evaluations and a novel error measurement. Compared with those of the other models, the final empirical results demonstrate that the proposed forecasting model can enhance the accuracy of crude oil futures forecasts.

1. Introduction

Crude oil is a crucial global fuel source and plays an extensive range of roles and functions in social and economic development (Abiad & Qureshi, 2023). Severe fluctuations in crude oil prices can influence other financial markets, including but not limited to the energy market and stock market (Mensi et al., 2023; Naeem et al., 2022). Accurate prediction of crude oil futures price movement is essential for macroeconomic policy making, financial derivative pricing, and risk management. Owing to the influence of various uncertain factors, the precision improvement of crude oil price remains challenging task (Guo et al., 2023; Li et al., 2022). Driven by this issue, the primary objective of this study is to improve the methodology for predicting crude oil futures prices.

In recent years, artificial intelligence models have been widely used for time series forecasting, including support vector machines (SVM), artificial neural networks (ANNs) and deep learning models. Zendehboudi et al. (2018) have carried out a comprehensive review on the SVM models for wind and solar energy resources. Findings from their critical analysis suggest that hybrid support vector machine models can achieve high accuracies. ANNs have the ability to acquire information from patterns and detect concealed functional connections within a given dataset, even if these connections are unknown or challenging to identify. The back-propagation neural network (BPNN) is considered a fundamental type of ANNs. The primary idea of this approach is to employ forward propagation to generate forecast values and rectify errors. Wang and Wang (2015) has established an improved stochastic BPNN neural network, and has analyzed the volatility statistics characteristics of the Chinese stock market indices. Ma et al. (2017) improved a type of quantum-inspired double parallel feed-forward neural network, that combines quantum computation theory with a double-parallel feed-forward neural network (DPFNN). Yang and Wang (2021) proposed a hybrid neural network prediction model based on wavelet neural network (WNN), which has been tested in predicting the impact on global energy prices. The experimental results showed that the proposed approach exhibited good generalization ability. Traditional neural network models are fully connected, with information flowing from the input layer to the hidden layer, and then to the output layer. There are no connections between nodes in the same layer, and the propagation of the network is sequential. A recurrent neural network (RNN) is a neural network structure with self-feedback neurons, which can store and memorize previous information in the hidden layer, and then input it into the hidden layer of the current calculation. Wang and Wang (2016) improved the forecasting accuracy

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of crude oil price fluctuations by combining the stochastic time effective function with the RNN model. A robust hybrid hours-ahead gas consumption method by integrating wavelet transform, RNN-structured model and genetic algorithm was established, and the effectiveness of the model was verified from multiple perspectives, including graphical examination, mathematical error analysis and model comparison (Su et al., 2019). In contrast to typical neural network models, deep learning approaches consisting of several hidden layers of multi-layer perceptrons, which exhibit superior performance in handling complex nonlinear properties (Fan et al., 2021). Hochreiter and Schmidhuber (1997) introduced an efficient, gradient based deep learning method called long short-term memory (LSTM), which can solve complex, artificial long-time-lag tasks that had never been solved by previous recurrent network algorithms. Sabri and El Hassouni (2023) proposed an LSTM autoencoder for photovoltaic power forecasting. Experiments demonstrated that the model achieved a better forecasting accuracy, but the training hyper-parameters is time-consuming. To improve the accuracy of stock price index prediction, Niu et al. (2021) introduced a hybrid model based on LSTM network, and the improved model has shown an outstanding performance in financial index forecasting. Subsequently, a plenty of time series forecasting models based on deep learning methods have been proposed, for instance, the deep LSTM (DLTSM) (Gulmez, 2023), the bidirectional LSTM (Lin et al., 2022; Wang & Wang, 2020), and variant deep learning models (Semenoglou et al., 2023; Wang et al., 2022a).

The enhancement of model architecture has been persistently investigated to improve forecasting accuracy and produce superior predictive performance (Fang et al., 2023a; Guo et al., 2023; Li et al., 2021; Wang et al., 2024a; Zhao et al., 2024). Among these models, hybrid models based on decomposition and integration frameworks have been widely recognized (Guo et al., 2023; Purohit & Panigrahi, 2024; Wang et al., 2023a, 2024b). The principle of hybrid models is to divide the original data series into several subseries and allow forecasting models to effectively learn the nonlinear properties of fluctuations in each subseries. To improve the forecasting accuracy by minimizing the impact of past information on the weights training process, several hybrid models have been developed by integrating the random theory with deep learning neural networks (Wang & Wang, 2016, 2021; Zhang et al., 2024, 2020), and some hybrid frameworks are enhanced by employing the attention mechanism (Liu & Zhou, 2024; Niu et al., 2024; Xu et al., 2024). The importance of the random behavior of asset series is shown in a large number of studies (Yang & Wang, 2021; Zhang et al., 2024, 2020). The crude oil futures market is an important subsector of the capital market. Crude oil futures fluctuation is regarded as a random process to some extents (Zhang et al., 2024). However, in the existing research, few researchers have taken this essential factor into account in the construction of the forecasting model.

Commonly employed decomposition methods include wavelet transform (Wang & Wang, 2020, 2021), empirical mode decomposition (EMD) (Fang et al., 2023b), EMD-based decompositions (Li et al., 2024; Wang et al., 2022b), variational mode decomposition(VMD) (Li et al., 2021), and so on. For the wavelet transform, it is difficult to choose an appropriate wavelet basis function. In addition, during the process of signal decomposition false spectrum always appear, and the results need to be further filtered and processed. The EMD method also has limitations, which are manifested in the existence of modal aliasing and endpoint effects. Subsequently, EMD-based decompositions have been proposed, including ensemble EMD (EEMD) (Li et al., 2024), and improved complementary EEMD(ICEEMD) (Wang et al., 2022a). Although the improved methods (EEMD and ICEEMD) can avoid the shortcomings of modal aliasing to a certain extent, it will still create new limitations. EEMD avoids mixing phenomena by adding white noise, but after signal reconstruction, the white noise remains and cannot be entirely eliminated. The disadvantage of ICEEMD is the uncertainty in the number of modes generated during decomposition, and it only partially reduces the impact of residual noise (Zhao et al., 2024, 2021). VMD is a non-recursive signal decomposition method that can decompose the signal into intrinsic mode functions with limited bandwidths and iteratively search for the optimal solution of the variational mode (Dragomiretskiy & Zosso, 2013). Compared with EMD based method, VMD overcomes the shortcomings of endpoint effects and mode aliasing, and has a solid mathematical theoretical foundation. It can decompose non-stationary signals contained high complexity into stationary sub-modes with multiple frequency scales. Yuan and Che (2022) proposed a short-term ensemble forecasting model based on VMD decomposition and two groups of multi-step strategies, and systematically studied the applicability. Parri et al. (2024) established a novel hybrid methodology by utilizing the VMD technique to extract the time-varying features of the wind speed data. The results of the two experiments indicate that the proposed approach exhibited superior performance, resulting in a significant improvement across all the time intervals evaluated. However, the decomposition process of VMD depends on the selection of parameters K and α. Recently, some researchers have attempted to optimize VMD parameters to enhance the time series prediction performance (Wang et al., 2023a; Xin et al., 2022). Wang et al. (2022b) proposed an optimal decomposition method by incorporating VMD and Sample entropy(SE). The experimental results demonstrated that decomposed subsequences are extracted effectively by the optimal method, which can improve the accuracy and efficiency of prediction. Nevertheless, SE can only reflect the complexity of a single time series, so modes are extracted according to an approximate value of sample entropy.

Based on the above literature, there are still some research gaps in current hybrid forecasting models. First, an optimal paradigm for effectively decomposing crude oil futures series is still needed. Although the VMD method has achieved good decomposition results, the problem remains that parameters need to be selected subjectively. Second, few articles have considered stylized empirical facts of crude oil futures series when modeling, such as volatility clustering and stochastic features. Current studies on financial time series forecasting are restricted to the incorporation of random theory into BPNN (Wang & Wang, 2015), WNN (Yang & Wang, 2021), or a singular LSTM model (Wang & Wang, 2020). Research on integrating random theory into the hybrid decomposition and integration models has been limited. Third, in terms of the error evaluation for forecasting results, traditional error indicators such as mean absolute error and root mean square error are generally applied, with little further analysis of the predictive accuracy from other perspectives.

To extend the limitations of the existing decomposition approaches and conventional deep learning methods, the research objectives of this paper are mainly focus on the following aspects. First of all, optimize the decomposition method, enabling the original series to be better decomposed and eliminating the current limitations (such as endpoint effects, mode aliasing and the problem of parameter selection). In addition, propose an enhanced financial forecasting model that takes into account the random characteristics of crude oil future prices. This model can dynamically assign different weights to different parts of information, selectively amplify critical information, and suppress noise, thereby improving prediction accuracy. Finally, the error assessment method should be enriched to compare the effectiveness of different forecasting models.

By considering the study motivation above, this paper puts forward a novel ensemble forecasting framework for crude oil futures prices. The primary contributions of this study might be summarized as follows:

(a) A novel optimized decomposition technique (OVMD) is developed by introducing the cross fuzzy entropy (C-FuzzyEn) method, which can accurately decompose series with data complexity reducing and features. It can solve the modal aliasing problem encountered in EMD, and EMD-type methods, and avoid the subjectivity of artificially setting parameters K and α in VMD.

(b) Attention mechanism is utilized to enhance the efficiency of the forecasting framework. It can focus on the important information transmitted by the previous layer by changing the distribution of weights. By adding the attention layer to the last layer, the depth of the framework is increased, and the forecasting precision of the crude oil futures sequences can be improved.

(c) A new error correction algorithm based on the stochastic strength formula is introduced to rectify the training error. Owing to crude oil futures sequences having stylized empirical facts (Zhang et al., 2020), the stochastic strength formula can help effectively capture the characteristics of fluctuations by assigning different weights for historical data according to its time distance.

(d) By integrating the OVMD method, attention mechanism and the new error correction algorithm into the deep BiLSTM model, an innovation hybrid forecasting system named OVMD-att-SSF-DBiLSTM is constructed. Compared with those of other forecasting models, the empirical results demonstrate the superiority of the proposed model through different error evaluation methods and statistical tests.

(e) A novel error measurement (MGCS) has been developed to evaluate and contrast the forecasting results of different models. Different from the traditional error criterion, the MGCS method explains the gap between actual values and predictive values by measuring consistency and synchronicity. The experimental results reveal that the MGCS method can effectively demonstrate the forecasting accuracy of different models.

The outline of this research is organized as follows. Section 2 introduces the methodologies and provides the main framework of this research. Section 3 describes the evaluation method utilized in the empirical analysis. Section 4 presents the empirical analysis and the comparative results in detail. Section 5 summarizes the main conclusions of this study.

2. Methodology

In this section, a novel hybrid forecasting system is established to improve the forecasting accuracy of crude oil futures prices. The proposed forecasting framework integrates the OVMD method, the attention mechanism based deep bidirectional LSTM model and the error correction algorithm based on stochastic strength formula. The basic principle of each method employed in the proposed model is described as follows.

2.1. Optimized variational mode decomposition

2.1.1. Variational mode decomposition

VMD is an entirely non-recursive signal processing algorithm which was proposed by Konstantin and Dominique (Dragomiretskiy & Zosso, 2013). In VMD, it is assumed that the center frequency bandwidth of each component is limited and that the decomposition process can be transformed into an optimization problem. It is constructed using Wiener filtering, the Hilbert transform, frequency mixing and heterodyne demodulation methods. In VMD, a signal series $f(t)$ is decomposed into a finite number of intrinsic mode functions (IMFs). Each IMF is chosen to be its bandwidth in spectral domain, that is, each mode requires to be mostly compact around a center frequency ω_k. The VMD algorithm comprises two main steps: constructing variational problem and then solving variational problem. For signal $f(t)$, the constrained variational problem is obtained as follows:

$$\begin{cases} \min_{u_k,\omega_k} \left\{ \sum_{k=1}^{K} \left\| \partial_t \left[(\delta(t) + \frac{j}{\pi t}) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ \qquad\qquad s.t. \quad \sum_{k=1}^{K} u_k = f \end{cases} \tag{1}$$

where $\delta(t)$ implies the Dirac distribution, u_k and ω_k denote the kth mode with limited bandwidth and the corresponding central frequency, f is the raw signal series, K represents the number of decomposed modes.

The introduction of penalty factor α and Lagrangian multiplier λ enables the transformation of the constrained variational problem into an unconstrained one. The unconstrained variational problem can be expressed as

$$\begin{aligned} \mathcal{L}(\{u_k\},\{\omega_k\},\lambda) = \alpha \sum_{k=1}^{K} \left\| \partial_t \left[(\delta(t) + \frac{j}{\pi t}) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ + \left\| f(t) - \sum_k u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_k u_k(t) \right\rangle . \end{aligned} \tag{2}$$

The alternating direction multiplier method is applied to solve this unconstrained variational problem. The optimal problem of the Lagrange expression can be solved by iteratively updating u_k, ω_k, λ. The iterative equations are as follows

$$\hat{u}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i<k} \hat{u}_i^{n+1}(\omega) - \sum_{i>k} \hat{u}_i^{n}(\omega) + \frac{\hat{\lambda}^n(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k^n)^2} \tag{3}$$

$$\omega_k^{n+1} = \frac{\int_0^\infty \omega |\hat{u}_k^{n+1}(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k^{n+1}(\omega)|^2 d\omega} \tag{4}$$

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau(\hat{f}(\omega) - \sum_k |\hat{u}_k^{n+1}(\omega)|) \tag{5}$$

where n stands for the iteration numbers, $\hat{u}_k^{n+1}(\omega)$, $\hat{f}(\omega)$, $\hat{u}_i(\omega)$ and $\hat{\lambda}^{n+1}(\omega)$ are Fourier transform forms of $u_k^{n+1}(\omega)$, $f(\omega)$, $u_i(\omega)$ and $\lambda^{n+1}(\omega)$, respectively. The convergence criterion expression of the iterative process is as follows

$$\sum_k \frac{\|\hat{u}_k^{n+1} - \hat{u}_k^n\|_2^2}{\|\hat{u}_k^n\|_2^2} < \varepsilon \tag{6}$$

where ε is the convergence tolerance.

2.1.2. Cross-fuzzy entropy

Cross-fuzzy entropy (C-FuzzyEn) is proposed to avoid the boundary limitation of cross-approximate entropy and cross sample entropy (Ramirez-Parietti et al., 2021; Xie et al., 2010), replacing the Heaviside function by introducing a Gaussian function in the calculation of the similarity between two vectors. The C-FuzzyEn is less rely on record length in evaluating the degree of pattern regularity and asynchrony between two time series. The C-FuzzyEn value of two variables is large when they exhibit weak regularity and asynchrony. Moreover, the C-FuzzyEn values will correspondingly decrease as their asynchrony increases.

For time series $\{X(i), i = 1, 2, \ldots, N\}$, $\{Y(j), j = 1, 2, \ldots, N\}$, C-FuzzyEn is defined as follows:

Step 1: The m-dimensional vectors $X^m(i)$ and $Y^m(j)$ as comprised as follows

$$X^m(i) = \{x(i), x(i+1), \ldots, x(i+m-1)\} - \bar{x}(i), \quad 1 \le i \le N-m+1 \tag{7}$$

$$Y^m(j) = \{y(j), y(j+1), \ldots, y(j+m-1)\} - \bar{y}(j), \quad 1 \le j \le N-m+1 \tag{8}$$

where $\bar{x}(i)$ and $\bar{y}(j)$ are the averages of the series segment, that is,

$$\bar{x}(i) = \frac{1}{m} \sum_{s=0}^{m-1} x(i+s) \tag{9}$$

$$\bar{y}(j) = \frac{1}{m} \sum_{s=0}^{m-1} x(j+s). \tag{10}$$

Step 2: Calculate the distance d_{ij}^m between vectors $X^m(i)$ and $Y^m(j)$.

$$d_{ij}^m = d\left[X^m(i), Y^m(j)\right] = \max_{k \in (0, m-1)} |\, x(i+k) - \bar{x}(i) - (y(j+k) - (\bar{y})(j)) \,| \, . \tag{11}$$

Step 3: The synchrony degree $D_{ij}^m(n,r)$ between $X^m(i)$ and $Y^m(j)$ can be calculated through a fuzzy function $\mu(d_{ij}^m, n, r)$

$$D_{ij}^m(n,r) = \mu(d^m(i,j), n, r) \tag{12}$$

where $\mu(d_{ij}^m, n, r) = \exp(-(d_{ij}^m)^n/r)$.

Step 4: The C-FuzzyEn between two time series is defined as

$$C - FuzzyEn(m, n, r) = -\lim_{N\to\infty}(\ln(\varphi^{m+1}/\varphi^m)) \tag{13}$$

where, the function φ^m is defined as

$$\varphi^m(n, r) = \frac{1}{N-m}\sum_{i=1}^{N-m}\left(\frac{1}{N-m}\sum_{j=1}^{N-m} D_{ij}^m(n, r)\right) \tag{14}$$

Due to the length N of time series is finite, the C-FuzzyEn result can be denoted as

$$C - FuzzyEn(m, n, r) = \ln\varphi^{m+1}(n, r) - ln\varphi^m(n, r). \tag{15}$$

Since the mode number K and penalty factor α in the VMD algorithm have a significant impact on the decomposition results, C-FuzzyEn is utilized to find the optimal K and α scientifically. The key factor of the proposed OVMD is to establish a rational fitness function, which can help find the optimal solution for the objective function during the global search process. The objective function is defined as:

$$\langle K, \alpha\rangle = \arg\min\left\{\frac{(k-2)!}{k!}\sum_i\sum_j CFE(u_i, u_j)\right\} \tag{16}$$

where $CFE(u_i, u_j)$ is the C-FuzzyEn value between different modes. to clearly express the implementation process of the proposed OVMD method, the pseudo-code is given as follows.

Algorithm 1. Optimize the VMD parameters by C-FuzzyEn

Input: Load the original time series $y = f(t)$
Step 1: Initial model parameters: number of modes K, penalty factor α, fitness value f_θ
Step 2: Optimized the parameters.
for $k = 1, 2, \dots, n$, $\alpha_0 = 100, 200, \dots, L$,
$u_k = VMD(y, K, \alpha)$
$f = \min\left\{\frac{(k-2)!}{k!}\sum_i\sum_j CFE(u_i, u_j)\right\}$
if $f < f_\theta$
$f_\theta = f$, $K = k$, $\alpha = \alpha_0$
end if
end for
Step 3: Output the optimal K and α.

2.2. The random deep bidirectional LSTM model

2.2.1. Deep bidirectional LSTM model

LSTM network is a particular form of RNNs that can handle long-term and short-term dependencies. LSTM was introduced by Hochreiter and Schmidhuber (1997). Based on conventional RNNs, cell states and gate mechanisms are added to the hidden layer so that the gradient vanishing problem can be mitigated mainly through control gates. In addition, each time the historical message is dispatched to the hidden layer's neurons, several control gates with different functions are employed to regulate the past and the latest information. The LSTM control gates involve three gates: the forget gate f_t, the input gate i_t and the output gate o_t. The input gate is used to determine how much current information should be retained in the cell state. The forget gate provides the channel to discard the useless information. The output gate regulates the final output. The main algorithm of LSTM model is expressed as follows (Niu et al., 2021; Wang & Wang, 2020).

$$f_t = \sigma\{W_f \cdot (x_t, h_{t-1}) + b_f\}. \tag{17}$$

$$i_t = \sigma\{W_i \cdot (x_t, h_{t-1}) + b_i\}. \tag{18}$$

$$\tilde{C}_t = tanh\{W_C \cdot (x_t, h_{t-1}) + b_C\}. \tag{19}$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t. \tag{20}$$

$$o_t = \sigma\{W_o \cdot (x_t, h_{t-1}) + b_o\}, \tag{21}$$

$$h_t = o_t \otimes tanh(C_{t-1}). \tag{22}$$

In the above equations, the following notation is used:

- x_t is the input vector at current time step t.
- W_f, W_i, W_C, W_o are the weight matrices which associate with corresponding vectors. They can be split into

$$\begin{cases} W_f = W_{fx} + W_{fh} \\ W_i = W_{ix} + W_{ih} \\ W_C = W_{Cx} + W_{Ch} \\ W_o = W_{ox} + W_{oh} \end{cases} \tag{23}$$

- b_f, b_i, b_C, b_o are bias indicators.
- f_t, i_t and o_t are forget gate, input gate and output gate vectors.
- C_t and $\tilde{C}_t$ are vectors for the cell states and candidate values, $\otimes$ denoting the Hadamard (element wise) product.
- h_t is a vector for the output of the LSTM layer.
- $\sigma(\cdot)$ and $tanh(\cdot)$ are the sigmoid function and hyperbolic tangent function, respectively.

Studies have revealed that increasing the depth of the network can effectively improve the training accuracy of the model (Niu et al., 2021; Wang & Wang, 2020). In this paper, a deep bidirectional Long short term memory network (DBiLSTM) is developed to predict the crude oil futures series. The framework of the DBiLSTM model consists of a bidirectional Long short term memory(BiLSTM) block, followed by two stacked layers of original LSTM. The first BiLSTM block is to correct and extract effective information, which can fully obtain the information provided by the input variables (Lin et al., 2022; Wang et al., 2023b). The output of the BiLSTM block serves as the input of the third LSTM layer. The topology of the DBiLSTM is shown in Fig. 1.

2.2.2. Error correction algorithm based on stochastic strength formula

In this section, the stochastic strength formula (SSF) based on geometric Brownian motion is utilized to carry out model error correction. By combining DBiLSTM model with error correction algorithm based on the SSF, the random DBiLSM model, named SSF-DBiLSTM, is established. Research reveals that messages in different past periods have varying effects on future forecasting results (Dufera, 2024; Dufresne, 2001). Specifically, the latest historical information is more influential than that from earlier periods. Given the different influence of historical data from different periods for forecasting future price trends, a new random effective function is proposed, which can assign different weights to different data in the light of the variant time of occurrence. The mathematical expression of the SSF is as follows.

$$\varphi(t_n) = \frac{1}{\beta}\exp\left\{-\int_{t_n}^{t_0}\mu(t)dt - \int_{t_n}^{t_0}\sigma(t)dB(t)\right\} \tag{24}$$

where $\beta(>0)$ represents the time intensity parameter, t_0 stands for the moment of the latest time point in the training set, and t_n is any time point in the training set. $B(t)$ is the standard Brownian motion. $\mu(t)$ is the drift function that mainly controls the changes in the overall trends, $\sigma(t)$ is the wave function, and its primary function is to model uncertain events during the forecasting process. The mathematical expressions of $\mu(t)$ and $\sigma(t)$ are as follows.

$$\mu(t) = exp(-\alpha t), \tag{25}$$

$$\sigma(t) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}. \tag{26}$$

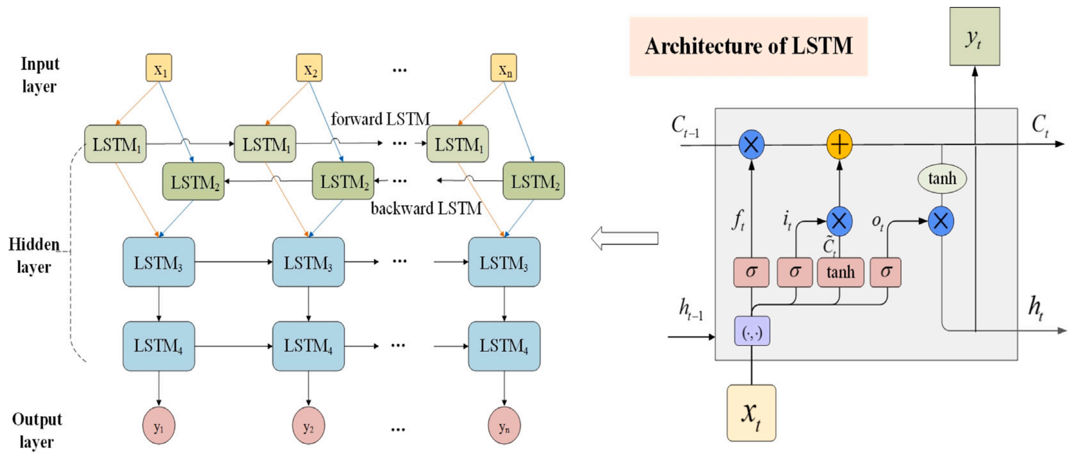


Fig. 1. The topology of DBiLSTM model.

In this section, we optimize the error correction process of model training by introducing the SSF method, which can optimize the parameter iteration efficiency, thereby enhancing the forecasting performance. Take the LSTM model as an example, and the detailed algorithm is as follows.

In the training process of the LSTM model, the parameter matrices W_f, W_i, W_C, and W_o are modified the back-propagation in each iteration through the time procedure of typical RNNs (Berradi & Lazaar, 2019). The model training error of the sample point n is defined as

$$E\left(t_n\right) = \frac{1}{2}\varepsilon_{t_n}^2 = \frac{1}{2}(d_{t_n} - y_{t_n})^2. \tag{27}$$

Combined with the error correction algorithm based on the SSF, the LSTM model training error E_{tn} is calculated as,

$$E\left(t_n\right) = \frac{1}{2}\varphi(t)\varepsilon_{t_n}^2 = \frac{1}{2}\varphi(t)(d_{t_n} - y_{t_n})^2. \tag{28}$$

The corresponding global error of model training is defined as

$$E = \frac{1}{N}\sum_{i=1}^{N} E_t = \frac{1}{2N}\sum_{i=1}^{N}\frac{1}{\beta}\exp\left\{\int_{t_n}^{t_0} -\mu(t)dt - \int_{t_n}^{t_0}\sigma(t)dB(t)\right\}\left(d_{t_n} - y_{t_n}\right)^2. \tag{29}$$

In the modeling process, based on the novel defined global error E, the model parameters are updated through the gradient descent method (Wang & Wang, 2021; Yang & Wang, 2021; Zhang et al., 2020). First, the partial derivative of each model parameter needs to be calculated from the global error function. The principle of parameter update is as follows.

$$\frac{\partial E}{\partial W_{fx}} = \frac{\partial E}{\partial net_{f,t}}\frac{\partial net_{f,t}}{\partial W_{fx}} = \delta_{f,t}\varphi(t)x_t, \tag{30}$$

$$\frac{\partial E}{\partial W_{fh}} = \sum_{j=t_0}^{t_n}\delta_{f,j}\varphi(t)h_{j-1}, \tag{31}$$

$$\frac{\partial E}{\partial W_{ix}} = \frac{\partial E}{\partial net_{i,t}}\frac{\partial net_{i,t}}{\partial W_{ix}} = \delta_{i,t}\varphi(t)x_t, \tag{32}$$

$$\frac{\partial E}{\partial W_{ih}} = \sum_{j=t_0}^{t_n}\delta_{i,j}\varphi(t)h_{j-1}, \tag{33}$$

$$\frac{\partial E}{\partial W_{Cx}} = \frac{\partial E}{\partial net_{C,t}}\frac{\partial net_{C,t}}{\partial W_{Cx}} = \delta_{C,t}\varphi(t)x_t, \tag{34}$$

$$\frac{\partial E}{\partial W_{Ch}} = \sum_{j=t_0}^{t_n}\delta_{C,j}\varphi(t)h_{j-1}, \tag{35}$$

$$\frac{\partial E}{\partial W_{ox}} = \frac{\partial E}{\partial net_{o,t}}\frac{\partial net_{o,t}}{\partial W_{ox}} = \delta_{o,t}\varphi(t)x_t, \tag{36}$$

$$\frac{\partial E}{\partial W_{oh}} = \sum_{j=t_0}^{t_n}\delta_{o,j}\varphi(t)h_{j-1}, \tag{37}$$

where $net_{f,t}$, $net_{i,t}$, $net_{C,t}$, $net_{o,t}$ denotes the input of the corresponding function, $\delta_{f,t} = \frac{\partial E}{\partial net_{f,t}}$, $\delta_{i,t} = \frac{\partial E}{\partial net_{i,t}}$, $\delta_{C,t} = \frac{\partial E}{\partial net_{C,t}}$, $\delta_{o,t} = \frac{\partial E}{\partial net_{o,t}}$.

2.3. Attention mechanism

Attention mechanism is an encoder–decoder type of neural network structure originated from the human visual mechanism simulation (Lin et al., 2022; Niu et al., 2024, 2021). When processing much complex information, human vision can fast pay attention to the essential areas and ignore the irrelevant information, then obtained the effective messages. Similarly, the attention mechanism filters valuable information and reduces the interference of worthless information by assigning different weights in the input sequence features, thereby reasonably focusing on more effective information, and improving model efficiency and accuracy (Niu et al., 2021; Wang et al., 2022a). The attention mechanism is applied as the last layer of deep learning models in this study. For instance, the att-SSF-DBiLSTM algorithm is established on the basis of the SSF-DBiLSTM model and includes the attention mechanism as the last layer of the model to help convert effective information into output variables. The attention mechanism layer can assign different weights to different parts of information dynamically, selectively amplify critical information, and suppress noise, thereby improving the forecasting accuracy of the output layer.

The process of attention mechanism is generally divided into three steps. Firstly, the similarity or correlation between the Query (output feature) and each Key (input feature) is calculated as follows:

$$s_t = \tanh(W_h h_t + b_h) \tag{38}$$

where h_t is the input vector, s_t is represents the importance of h_t, W_h and b_h are the weight and bias values, respectively. In the second step, the softmax function is employed to convert attention score into the probability form.

$$a_t = softmax(s_t) = \frac{\exp(s_t)}{\sum_j \exp(s_t)} \tag{39}$$

At last step, the final attention value is calculated by the weighted summation of values as follows:

$$z = \sum_t a_t h_t. \tag{40}$$

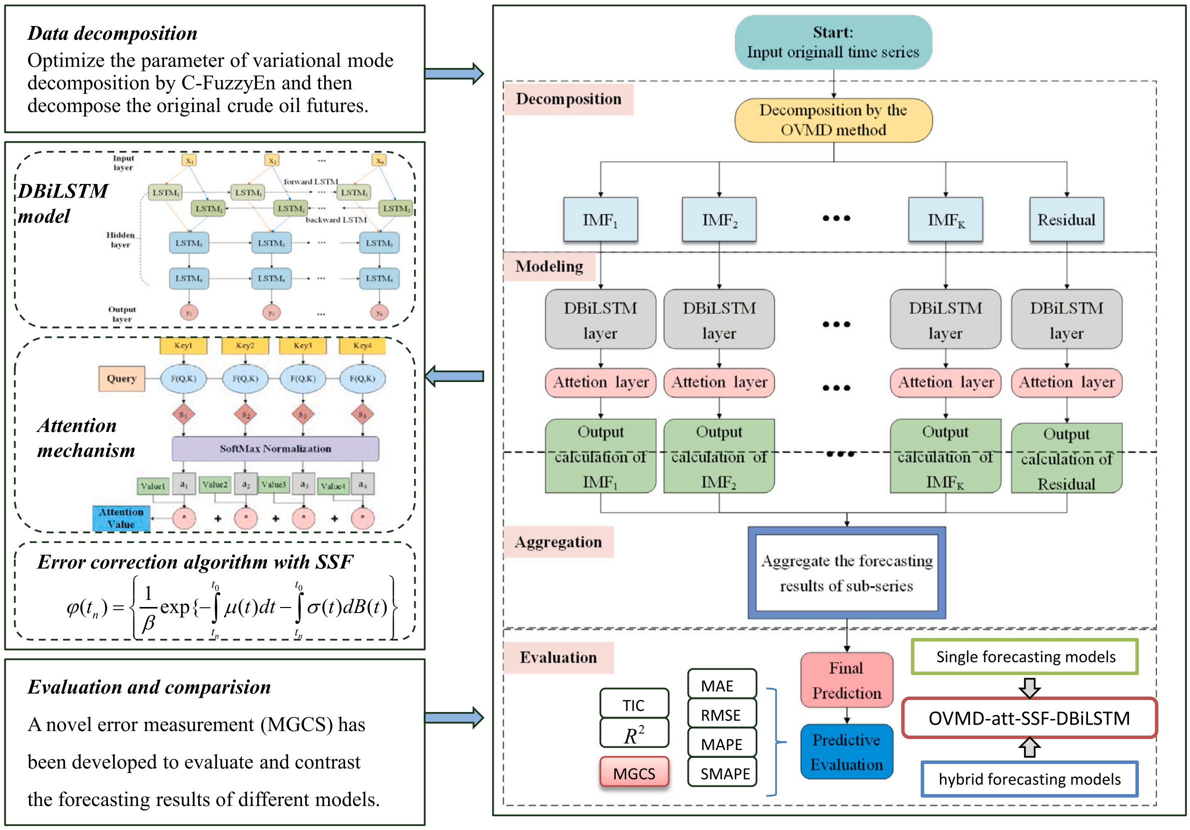


Fig. 2. Flowchart of the proposed OVMD-att-SSF-DBiLSTM.

2.4. The proposed OVMD-att-SSF-DBiLSTM algorithm

Based on the three theoretical innovations mentioned above, the novel hybrid system called OVMD-att-SSF-DBiLSTM has been proposed for crude oil futures sequences forecasting. The main procedure of the proposed hybrid forecasting system is shown in Fig. 2. The key steps of the algorithm are introduced as follow:

Step 1: The OVMD method is applied to determine the mode number K and penalty factor α of VMD.

Step 2: The OVMD technique is employed to decompose the original crude oil futures series $X(t)$ $(t = 1, 2, \ldots, N)$. Then intrinsic mode functions $IMF_i, i = 1, 2, \ldots, K$ are derived.

Step 3: Each IMF_i decomposed from the OVMD method is divided into training set and testing set. Then the att-SSF-DBiLSTM network is utilized for training and establishing K forecasting sub-models.

Step 4: Calculate the predictive values of all the sub-models with the well-trained models.

Step 5: Composite the prediction of each IMF_i to obtain the final forecasting results for the crude oil futures series. Moreover, linear regression and relative error are applied to investigate the correlation between the predictive points and actual values.

Multiple evaluation indicators are adopted to estimate the prediction effectiveness of OVMD-att-SSF-DBiLSTM, including MAE, RMSE, MAPE, SMAPE and TIC, and the newly proposed error measurement (MGCS). In addition, eleven forecasting models are viewed as benchmark models for empirical error analysis and comparison.

3. Evaluation methods

3.1. Performance evaluation with statistical methods

To estimate the forecasting error of the new hybrid model and compare it with that of five other models, the error measurement between actual data points and predicted values for different models are investigated. Among them, mean absolute error (MAE), root mean square error (RMSE), mean absolute percent error (MAPE), symmetric mean absolute percent error (SMAPE) and the Theil inequality coefficient (TIC) are selected as the error evaluation criteria to indicate the forecasting performance of each model. Generally, the smaller the error (MAE, RMSE, MAPE, SMAPE and TIC) values are, the more accurate the predictive ability of the forecasting model (Makridakis, 1993). The evaluation definitions are expressed as follows.

$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N}|y_t - \hat{y}_t|, \tag{41}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}, \tag{42}$$

$$\mathrm{MAPE} = 100 \times \frac{1}{N}\sum_{t=1}^{N}\left|\frac{y_t - \hat{y}_t}{y_t}\right|, \tag{43}$$

$$\mathrm{SMAPE} = 100 \times \frac{2}{N}\sum_{t=1}^{N}\frac{|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|}, \tag{44}$$

$$\mathrm{TIC} = \frac{\sqrt{\frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}}{\sqrt{\frac{1}{N}\sum_{t=1}^{N}{y_t}^2} + \sqrt{\frac{1}{N}\sum_{t=1}^{N}\hat{y}_t^2}}, \tag{45}$$

where y_t and $\hat{y}_t$ are the actual and the predicted values at time t, respectively. N is the total number of the data.

The Diebold–Mariano(DM) test, a hypothesis testing method, is adopted to verify the significant differences between the proposed model and the other compared forecasting models. This test measures the different degrees of forecasting precision among models from a statistical perspective (Diebold & Mariano, 2002). The Wilcoxon signed-rank test is a nonparametric test method used to compare the differences between two sets of paired data (Rosner et al., 2006). In order to further compare the differences between the proposed model and the benchmark models, the Wilcoxon signed-rank test is conducted on the prediction results of the comparison models. It is commonly known that the predicted results and the actual values can be fitted via the linear regression method. Through establishing a linear regression equation $Y = aX + b$, the accuracy of the forecasting model can be reflected. The closer to 1 the coefficient a value is, the closer the predicted value is to the true value, and the more prominent the model prediction effect. Moreover, the accuracy of the forecasting models can be judged via the fitting goodness index R^2.

3.2. Methodology of MGCS analysis

Multiscale generalized complexity synchronization (MGCS) analysis is based on the generalized multiscale complexity-invariant distance(GMCID) and the multiscale sample entropy (Armand Eyebe Fouda, 2016; Gao et al., 2019). The novel MGCS analysis method can evaluate the predicted performance between two financial time series. GMCID is defined by improving the conventional complexity-invariant distance analysis with the Minkowski distance to measure the differences between two time series (Batista et al., 2014) Sample entropy, as a method to measure the degree of uncertainty, has a wide application in detecting system complexity of financial time series (Richman et al., 2004). First, based on the improved formalism of the Minkowski distance, the generalized CID (GCID) between the time series x and y is defined as

$$GCID(x, y, q) = \|x - y\|_q^q \times \frac{max\{\| \triangle x\|_q^q, \| \triangle y\|_q^q\}}{min\{\| \triangle x\|_q^q, \| \triangle y\|_q^q\}} \tag{46}$$

where $\|x\|_q^q = \sum_{i=1}^n |x_i|^q, \triangle x = \{x_2 - x_1, x_3 - x_2, \dots, x_n - x_{n-1}\}, |x - y|^q = \{|x_1 - y_1|^q, |x_2 - y_2|^q, \dots, |x_n - y_n|^q\}$.

Second, we integrate the sample entropy and complexity-invariant distance into MGCS to study the multiscale structure of the fluctuation trends between time series. Consider time series $x_t, t = 1, 2, \dots, n$, where n is the length of the time series. For an embedding dimension m, the m-dimensional sequence vectors can be defined as

$$x_m(i) = \{x_{i+k} : 0 \leqslant k \leqslant m - 1\} \quad 1 \leqslant i \leqslant n - m + 1, \tag{47}$$

The distances between the two vectors $x_m(i)$ and $x_m(j)$ are considered as

$$d(x_m(i), x_m(j)) = max\{|x_{i+k} - x_{j+k}| : 0 \leqslant k \leqslant m - 1\},$$

where $i, j = 1, 2, \dots, n - m + 1$. Given tolerance $\hat{r}$ in advance, if $d(x_m(i), x_m(j))$ is smaller than $\hat{r}$, indicating that the two vectors are similar. Here, $\hat{r} = k * SD(0.1 \leq k \leq 0.25)$, where SD is the standard deviation of the actual value. We calculate the MGCS with the parameter $\hat{r} = 0.15 * SD$. Then let $C_i^m(\hat{r})$ represent the number of vectors $x_m(i)$ whose distance of $x_m(j)$ is smaller than a specified tolerance level $\hat{r}$ while excluding self-matches. Similarly, $C_i^{(m+1)}$ presents the number of matches of length $m+1$. The sample entropy of time series x is defined as follows

$$SampEn(x, m, \hat{r}) = -\ln\left\{\frac{\sum_{i=1}^{n-m} C_i^{(m+1)}(x, \hat{r})}{\sum_{i=1}^{n-m+1} C_i^{(m+1)}(x, \hat{r})}\right\}. \tag{48}$$

Third, before quantifying the difference between the time series pairs, a coarse-grained approach should be applied to the original time series. The coarse-grained time series of x and y in the scale factor τ is computed as follows

$$x_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^{j\tau} x_i, \quad 1 \leq j \leq N/\tau, \tag{49}$$

$$y_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^{j\tau} x_i, \quad 1 \leq j \leq N/\tau. \tag{50}$$

where s represents the length of each coarse-grained time series and is equal to the quotient of N/τ.

Table 1
Data selection.

Index	Data sets	Date	Number
Brent	Training set	02/01/2011 ∼ 21/09/2018	1995
	Validation set	24/09/2018 ∼ 22/04/2021	665
	Testing set	23/04/2021 ∼ 15/11/2023	665
WTI	Training set	02/01/2011 ∼ 11/10/2018	2019
	Validation set	12/10/2018 ∼ 05/05/2021	673
	Testing set	06/05/2021 ∼ 15/11/2023	673

Then MGCS between x and y is defined as

$$MGCS(x, y, q, \tau) = \left\{GCID(x^{(\tau)}, y^{(\tau)}, q) \times SampEn(|x^{(\tau)} - y^{(\tau)}|^q, m, \hat{r})\right\}^{\frac{1}{q}}. \tag{51}$$

By introducing the GCID method into the sample entropy, MGCS can characterize the distinctions between two time series in terms of the entropy and distance complexity. The novel error measurement considers multiple time scales when the correlations between different futures series are validated and quantified.

4. Experimental analysis

4.1. Data selection and preprocessing

This research mainly focuses on the crude oil futures market, where the Brent and the West Texas Intermediate (WTI) crude oil futures price series are selected for the case study. Brent crude oil is the benchmark for crude oil in Africa, Europe and the Middle East, which is often treated as the standard of the organization of the petroleum exporting countries. WTI crude oil is produced in the United States and is regarded as the benchmark for crude oil in the Western Hemisphere. The datasets can be downloaded from https://cn.investing.com/. Table 1 displays the selected Brent and WTI crude oil data sets from 02/01/2011 to 15/11/2023. Usually, non-trading days are regarded as frozen such that only the data during trading time are adopted in this study. In the experimental process, 60% of the sample points are selected as the training set, the next 20% serve as the validation set and the remaining 20% of the data are utilized for testing and verifying the predictive ability of the models. Table 1 shows the specific data selection of the two indices.

Generally, to minimize the influence of noise and finally enhance the forecasting accuracy, the given data is normalized to the range of $[0, 1]$ by the following standardized method (Lin et al., 2022; Makridakis, 1993):

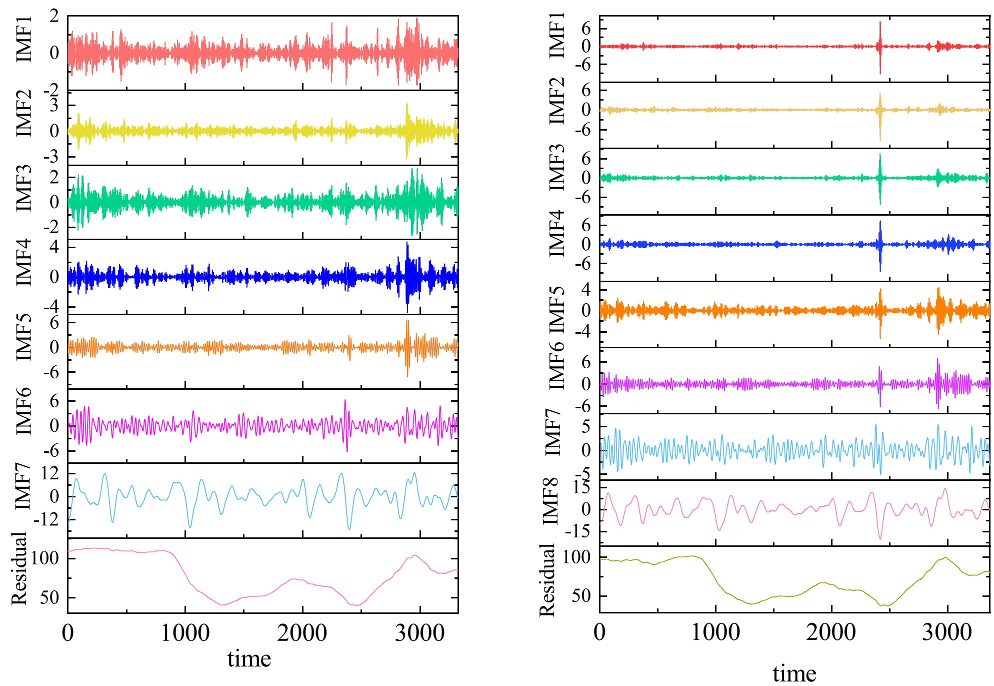
$$S(t)' = \frac{S(t) - \min S(t)}{\max S(t) - \min S(t)} \tag{52}$$

Afterward, to acquire accurate predictive values and make direct comparisons between the numerical results and the actual values the normalized output variables $S(t)'$ should be converted back to $S(t)$ using the following formula,

$$S(t) = S(t)'(\max S(t) - \min S(t)) + \min S(t). \tag{53}$$

4.2. Training and forecasting by proposed model

In this section, Brent and WTI crude oil futures sequences are conducted to bolster the proposed hybrid OVMD-att-SSF-DBiLSTM model. To begin with, by applying the OVMD technique, the Brent futures series is divided into 8 sub-modes, labeled $IMF_i, i = 1, 2, \dots, 7$, along with a residual component, and the WTI futures series is decomposed into 9 sub-series, denoting as $IMF_i, i = 1, 2, \dots, 8$ and a residual



(a) Brent (b) WTI

Fig. 3. (a) IMF components and the residual for Brent futures sequence. (b) IMF components and the residual for WTI futures sequence.

component. The advantage of this decomposition method is that the parameters of the VMD can be determined, and the drawbacks of EMD-type decomposition methods are avoided. Fig. 3 presents the sub-mode components of the two futures series decomposition. Due to this research focuses on the innovation of the model framework. To avoid the impact of different input variables on model accuracy, 4 classic variables with a moving window size of 5 are selected during the training phase, including the closing price, the opening price, the maximum price and the minimum price. The size of the moving window is set based on 5 trading days per week, which can be simply regarded as a weekly cycle. The hyper parameters of the proposed model are chosen by calculating the root mean square error between the predicted results and actual values (Wang & Wang, 2020; Yang & Wang, 2021). After extensive empirical experiments, the hyper parameters with the lowest root mean square error value are chosen as the optimal parameters. In the proposed model, the batch size is 64, the hidden size equals 35 and the epochs are set to 300.

Afterwards, the normalized subseries IMF_i and residual obtained from OVMD are trained and predicted by the att-SSF-DBiLSTM model. This stage is an essential component that assesses the prediction fluctuations, particularly in precisely forecasting the fluctuations direction. Fig. 4 shows the forecasting results of each subseries from the futures series of Brent and WTI crude oil. The graphic representation clearly shows that the predicted value of each subseries IMF_i matches precisely with the actual values. The residual subseries is acknowledged as the overall trend of the futures price series, whose results from the proposed forecasting model are well predicted, and curves of the actual data and the predicted data are intuitively quite approximative.

The final predictive outcomes for the two crude oil futures indices using the proposed model are depicted in Fig. 5. From this figure, the fluctuation trends of the predictive data are remarkably similar to those of the actual data, particularly for the testing set. It can be concluded that the behaviors of the predicted results nearly have consistent trends with the fluctuations of the actual data. Furthermore, the relative error outcomes of the empirical analysis are also illustrated in Fig. 5. These outcomes can be estimated by $RE(t) = |\hat{y}_t - y_t|/y_t$. The RE results for the Brent and WTI futures sequences are centralized in $(0, 0.4)$, and only a few points in periods of extreme volatility exceed 0.1. This finding indicates that the crude oil futures sequences have been trained exceptionally well through repeated experiments, and the forecasting performance of the proposed att-SSF-DBiLSTM model is worth approving.

4.3. Model comparison and prediction accuracy evaluation

While the proposed OVMD-att-SSF-DBiLSTM model is utilized in forecasting experiments, it is also essential to compare the predicted outcomes with those of other models. Two categories of forecasting models, namely single models and hybrid models, are employed as benchmark models. Six single models, SVM, BPNN, LSTM, DBiLSTM, SSF-DBiLSTM and SSF-att-DBiLSTM, are employed to compare forecasting evaluations. The SVM technique is regarded as state-of-the-art machine learning theory for binary classification. The BPNN is a type of traditional neural network. Since the proposed model is built upon the LSTM network, the LSTM, DBiLSTM, SSF-DBiLSTM and att-SSF-DBiLSTM models are chosen to demonstrate the predictive effects of the proposed model for thorough comparison. The frameworks of LSTM and DBiLSTM are given at Section 2.2.1. The principle of the SSF-DBiLSTM model integrates the error correction algorithm based on the SSF to the DBiLSTM model. The att-SSF-DBiLSTM model is an extension of the SSF-DBiLSTM model, incorporating an additional attention layer at the final layer. In addition, we selected five hybrid models, namely OVMD-att-DBiLSTM, OVMD-DBiLSTM, EEMD-DBiLSTM, OVMD-BPNN, and EEMD-BPNN, for comparison. The OVMD-att-DBiLSTM model combines the OVMD method with the attention mechanism layer as the final layer of the DBiLSTM model. The DBiLSTM model serves as the basis for both the OVMD-DBiLSTM and the EEMD-DBiLSTM, with the OVMD-DBiLSTM employing the OVMD method and the EEMD-DBiLSTM model adopting the EEMD method in the decomposition stage. Similarly, both the OVMD-BPNN and the EEMD-BPNN are based on the traditional BPNN model ; however, the OVMD-BPNN model utilizes the OVMD method during the decomposition stage, while the EEMD-BPNN model applies the EEMD method. In order to demonstrate

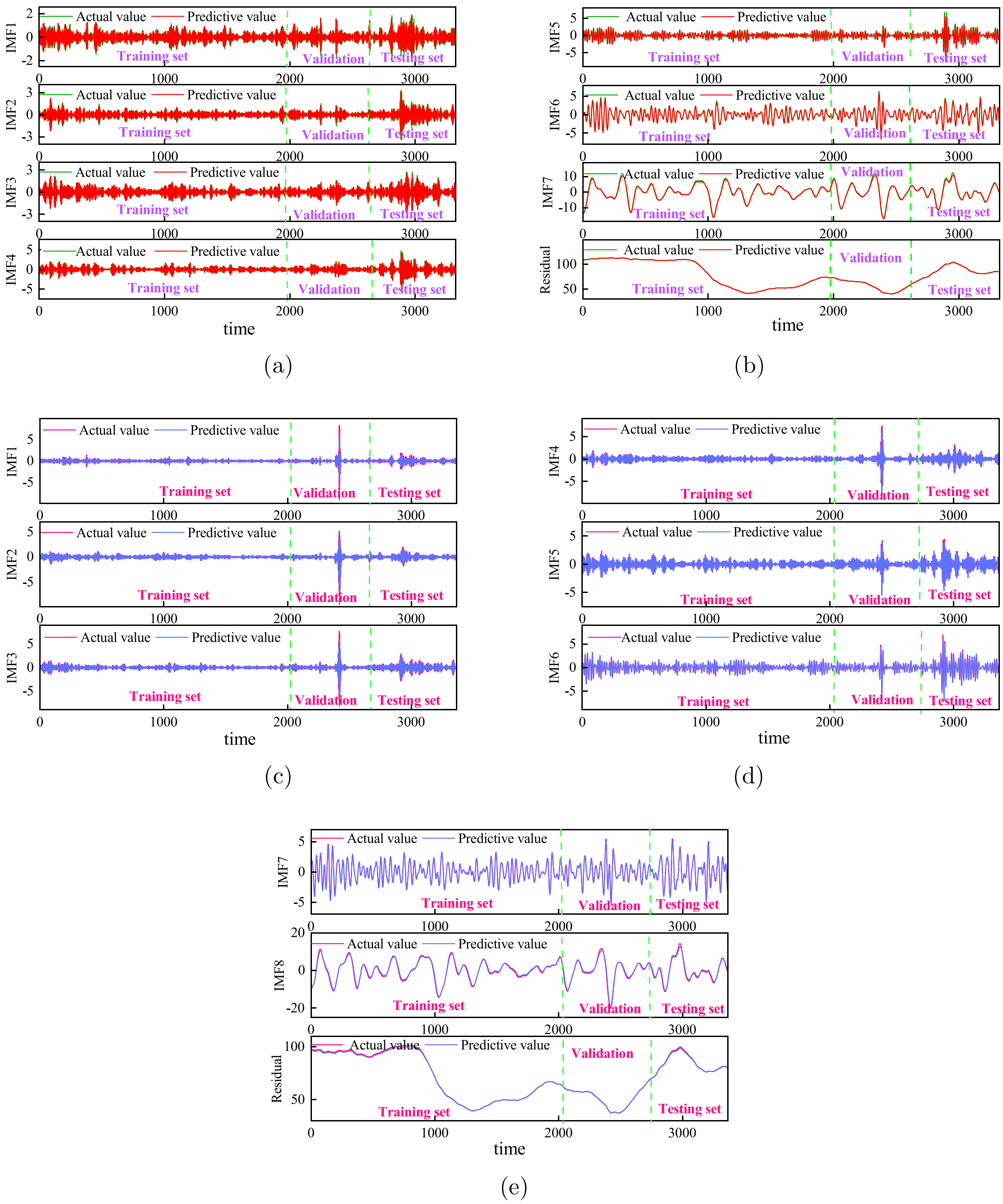


Fig. 4. (a) (b) The predicted data and the actual data of each sub-series from Brent crude oil futures prices. (c)(d)(e) The predicted data and the actual data of each sub-series from WTI crude oil futures prices.

the effectiveness of the innovations, input variables of all the comparative models are consistent with the inputs of the proposed model in the training process. The hyper parameter settings of all DBiLSTM-based hybrid models and the LSTM model are the same as those of the proposed model. In this study, the parameters are determined based on the results of repeated experiments. For the SVM model, the Gaussian kernel function is selected as the kernel function and the penalty coefficient $C = 100$ and $\gamma = 0.001$ is chosen because this set can produce the best possible results according to the validation set. Following the principles introduced in Wang and Wang (2015) and large amount of training experiments for the BPNN model and BPNN-based hybrid models, the neuron number of the input layer is set to 5, the number of the hidden layer is chosen as 9, the number of the output is determined to 1, the learning rate is 0.001 and the maximum training iterations number is selected to 300. For LSTM based models, the number of input units is set to 4, the number of hidden units is determined to 35, the number of the output unit is 1, the number of time steps is set to 5, the batch size is 64 and the epochs are 300. The specific parameter settings for forecasting models are listed in Table 2. All experiments were conducted in Python 3.9.7 running on the Windows 10 with 64-bit operating system. The CPU specification is Intel(R) Core(TM) i7-8750H processor and the GPU is NVIDIA Geforce GTX 1060 with 16 GB RAM.

Fig. 6 depicts the forecasting results of the Brent and WTI futures sequences for the compared forecasting models. Fig. 6(a) and (b) depict the comparison results of the proposed model and single models, and

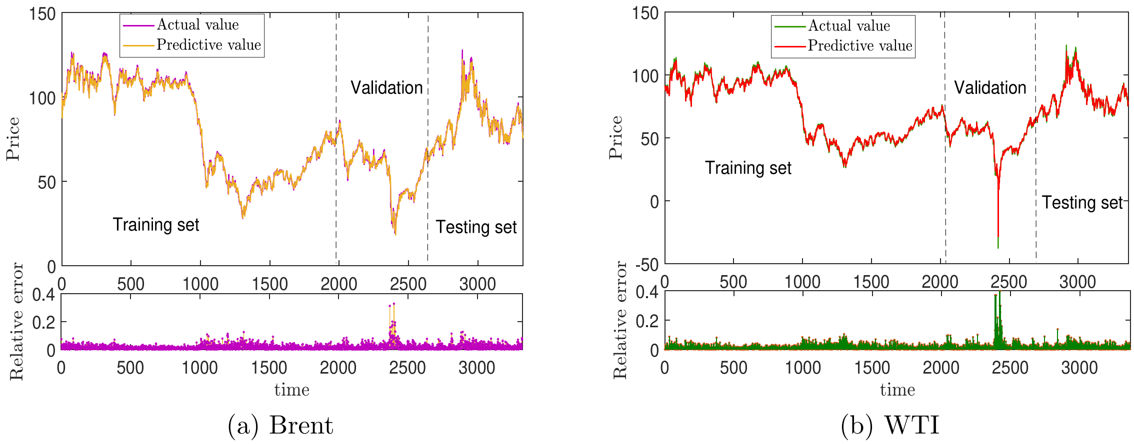


Fig. 5. The predicted results of the proposed model for Brent and WTI crude oil futures sequences.

Fig. 6(c) and (d) show the comparison between the proposed model and hybrid models. In contrast, Fig. 6(e) and (f) exhibit the outcomes of all comparative experiments. The forecasting results from the insert plots in Fig. 6 demonstrate the local prediction of both the training and testing sets from the forecasting models. The proposed model exhibits remarkable advantages over the other models, particularly during the sharp fluctuation periods. Moreover, hybrid models outperform single models in both forecasting fluctuations and trends. Influenced by changes of the socio-economic landscape and diverse external environments, the crude oil futures market appears different fluctuations. The predicted results during the minor fluctuations period appear to be relatively reliable for all forecasting models. Hybrid models exhibit enhanced prediction capabilities during extreme fluctuations periods.

Fig. 7 displays the relative error of prediction for the Brent and WTI futures sequences, allowing for a comparison of the forecasting results of different models. Comparing the relative error results of all the forecasting models, the relative error of the proposed model is the lowest and the distribution is the most concentrated. In addition, the requirement of the error correction algorithm based on the SSF is proven by comparing the prediction relative errors of two groups of models: the proposed OVMD-att-SSF-DBiLSTM model and the OVMD-att-DBiLSTM model, and SSF-DBiLSTM and DBiLSTM. The compared numerical results between the proposed OVMD-att-SSF-DBiLSTM model and att-SSF-DBiLSTM illustrate the importance of decomposition. Comparisons of OVMD-DBiLSTM and EEMD-DBiLSTM, OVMD-BPNN and EEMD-BPNN reveal the advantages of the OVMD method.

Linear regression curves for the Brent and WTI crude oil futures series are displayed in Figs. 8 and 9, while the corresponding statistical results can be found in Table 3. To be specific, the values of R^2 of the proposed OVMD-att-SSF-DBiLSTM model for the two futures series all exceed 0.99. Compared with the other forecasting models, the R^2 of the proposed model is closest to 1. Simultaneously, when compared to the other forecasting models, the regression coefficients a of the proposed model for the two crude oil series are nearest to 1, which indicates that the predicted values closely approximate the actual values. The regression equation parameters of the proposed model for the Brent crude oil futures series are $a = 0.9927$ and $b = 1.0950$, which is approaching to the ideal situation $y = x$, followed by the WTI futures indices, with parameter values of $a = 0.9789$ and $b = 2.1253$.

Additionally, for the Brent crude oil futures, the a value of the att-SSF-DBiLSTM model is 0.9781, and the R^2 value is 0.9935. The fitting results of the att-SSF-DBiLSTM model is not as satisfactory as that of the proposed model, which indicates that the OVMD method can help improve the prediction accuracy. For the Brent futures predicted results of the OVMD-att-DBiLSTM model, the fitting parameter a is 0.9791, and R^2 is 0.9937, which is worse than that of the proposed model. The comparison results demonstrate the advantages of the error correction algorithm based on the SSF. When the fitting results of OVMD-DBiLSTM and EEMD-DBiLSTM, as well as OVMD-BPNN and EEMD-BPNN are compared, it is evident that the OVMD-based hybrid forecasting model significantly improves the prediction accuracy. It should also be noted that, for all forecasting models, the forecasting accuracy of WTI is far inferior to that of Brent. The main reason is that WTI crude oil futures exhibited sharp fluctuations in April 2020, particularly on April 20th, when the price plummeted to -37.63 per barrel. The numerical predicted results verify the prediction accuracy of the proposed model for extreme situations.

Tables 4 and 5 provide a comprehensive quantitative comparison of the evaluation criteria, including the MAE, RMSE, MAPE, SMAPE, and TIC, for the indicated forecasting models. The numerical results indicate that the evaluation criteria of the proposed OVMD-att-SSF-DBiLSTM model are consistently the lowest among all the models. Additionally, the evaluation criteria of the hybrid models are much lower than those of the corresponding single models. To be specific, the MAE value for Brent futures sequence from the proposed model is 1.3952, and the values of RMSE, MAPE, SMAPE, and TIC are 2.0404,1.5655,1.8134 and 0.0104, respectively. For att-SSF-DBiLSTM, the values of MAE, RMSE, MAPE, SMAPE, and TIC are 1.9644, 2.8703, 2.3201,2.7071 and 0.0171, respectively. The comparison findings demonstrate that the prediction accuracy can be enhanced by integrating the OVMD approach. This conclusion can also be obtained by comparing the evaluation criteria results of the two sets of model: OVMD-DBiLSTM and DBiLSTM, OVMD-BPNN and BPNN. Further comparison of the prediction results from the two groups of models OVMD-DBiLSTM and EEMD-DBiLSTM, OVMD-BPNN and EEMD-BPNN reveal that the OVMD-based model exhibits superior prediction accuracy, indicating that the OVMD decomposition approach proposed in this paper is achievable and efficient. For the OVMD-att-DBiLSTM model, the values of MAE, RMSE, MAPE, SMAPE, and TIC are 1.7077, 2.5744, 2.0176, 2.4053 and 0.015, respectively. It is evident that the accuracy of this model is inferior to the predictive accuracy of the proposed OVMD-att-SSF-DBiLSTM model. At the same time, comparing the prediction error results of the SSF-DBiLSTM model and the DBiLSTM model, we discover that the SSF-DBiLSTM model has better prediction results and better accuracy. Thus, it can be inferred that the forecasting model employing the error correction algorithm based on the SSF has the ability to significantly enhance the predictive precision of deep learning models. In addition, while comparing the prediction results of two pairs of models, namely OVMD-att-DBiLSTM and OVMD-DBiLSTM, att-SSF-DBiLSTM and SSF-DBiLSTM, the incorporation of the attention mechanism layer into the deep learning method increases the depth of the model. Consequently, the prediction model becomes more proficient at capturing crucial information while minimizing the impact of irrelevant data, thereby enhancing the prediction accuracy.

Table 2
Parameter settings of forecasting models.

Model	Hyper-parameter	Value
Proposed Model	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	4: a bidirectional training layer, follows two LSTM unit layers and a attention mechanism layer
OVMD-att-DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	4: a bidirectional training layer, follows two LSTM unit layers and a attention mechanism layer
OVMD-DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	3: a bidirectional training layer and two LSTM unit layers
EEMD-DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	3: a bidirectional training layer and two LSTM unit layers
OVMD-BPNN	Input units	5
	Hidden layer neurons	9
	Output units	1
	Learning rate	0.001
	Maximum training iterations	300
EEMD-BPNN	Input units	5
	Hidden layer neurons	9
	Output units	1
	Learning rate	0.001
	Maximum training iterations	300
att-SSF-DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	4: a bidirectional training layer, follows two LSTM unit layers and a attention mechanism layer
SSF-DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	3: a bidirectional training layer and two LSTM unit layers
DBiLSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	3 : a bidirectional training layer and two LSTM unit layers

(*continued on next page*)

Table 2 (*continued*).

Model	Hyper-parameter	Value
LSTM	Input units	4
	Hidden layer neurons	35
	Output units	1
	Number of time steps	5
	Batch size	64
	Training epochs	300
	Depth of training layer	1: a single LSTM layer
BPNN	Input units	5
	Hidden layer neurons	9
	Output units	1
	Learning rate	0.001
	Maximum training iterations	300
SVM	Kernel function	Gaussian kernel function
	Penalty coefficient	100
	γ	0.001

Table 3
Linear regression parameters from forecasting model.

Model	Brent			WTI		
	a	b	R^2	a	b	R^2
Proposed Model	0.9927	1.0950	0.9970	0.9789	2.1253	0.9927
OVMD-att-DBiLSTM	0.9791	1.3261	0.9937	0.9370	3.6844	0.9916
OVMD-DBiLSTM	0.9565	2.6812	0.9919	0.8730	6.2499	0.9892
EEMD-DBiLSTM	0.9489	3.5267	0.9910	0.8555	6.8850	0.9882
OVMD-BPNN	0.8826	6.6991	0.9903	0.7292	11.8048	0.9872
EEMD-BPNN	0.8819	5.9283	0.9898	0.7011	11.0825	0.9864
att-SSF-DBiLSTM	0.9781	1.4252	0.9935	0.9358	4.7240	0.9909
SSF-DBiLSTM	0.9509	4.1348	0.9928	0.8863	5.6262	0.9885
DBiLSTM	0.9318	6.2726	0.9915	0.8733	6.2017	0.9870
LSTM	0.9212	4.2677	0.9910	0.7822	9.8381	0.9861
BPNN	0.8570	8.1001	0.9895	0.6295	15.9461	0.9849
SVM	0.8456	8.4364	0.9881	0.5960	16.9585	0.9835

Table 4
Prediction performance evaluation of predictive models for Brent crude oil futures.

Model	MAE	RMSE	MAPE	SMAPE	TIC
Proposed model	1.3952	2.0404	1.5655	1.8134	0.0104
OVMD-att-DBiLSTM	1.7077	2.5744	2.0176	2.4053	0.0152
OVMD-DBiLSTM	1.9157	2.7830	2.2269	2.6433	0.0165
EEMD-DBiLSTM	2.1281	3.0296	2.5085	2.9114	0.0196
OVMD-BPNN	2.9912	4.1999	3.2428	3.7254	0.0248
EEMD-BPNN	3.5949	4.7063	3.9412	4.4066	0.0279
att-SSF-DBiLSTM	1.9644	2.8703	2.3201	2.7071	0.0171
SSF-DBiLSTM	2.0630	3.0100	2.4635	2.9662	0.0187
DBiLSTM	2.3251	3.3543	3.0019	3.1717	0.0213
LSTM	2.5610	3.6821	4.0024	3.5393	0.0239
BPNN	3.7307	4.7929	5.2539	5.0073	0.0284
SVM	4.1029	5.2708	5.6185	5.3604	0.0313

By assessing the predicted evaluations of the WTI futures sequence, the aforementioned conclusion remains valid.

For the sake of improving the graphic description of the forecasting results, Figs. 10 and 11 present the evaluation values of the Brent and the WTI futures sequences, including the MAE, RMSE, MAPE, SMAPE, and TIC for the proposed model and benchmark models. As depicted in the figures, the prediction accuracy of the proposed model is the highest. Meanwhile, the prediction accuracy of the WTI futures sequence is not as high as that of the Brent futures sequence, and there exists a substantial disparity between the forecasting models. Specifically, the conventional BPNN, SVM, hybrid OVMD-BPNN, and EEMD-BPNN models yield inadequate predictions for the WTI futures sequence. Because there are extreme fluctuations in the WTI futures sequence, the findings indicate that the proposed model is better at predicting extreme fluctuations than the comparative models are.

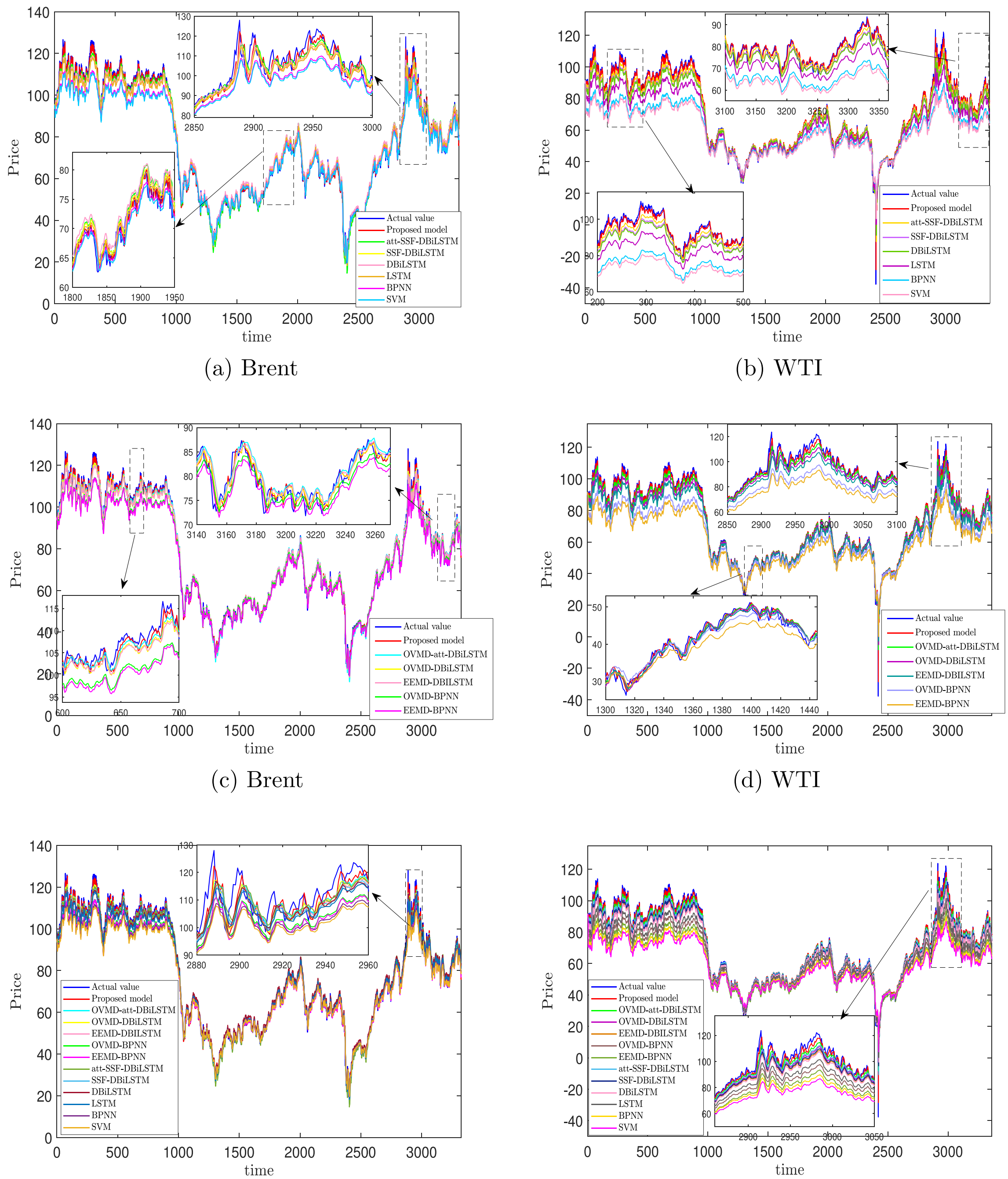
Table 5
Prediction performance evaluation of predictive models for WTI crude oil futures.

Model	MAE	RMSE	MAPE	SMAPE	TIC
Proposed model	1.5667	2.2322	1.8659	1.8665	0.0137
OVMD-att-DBiLSTM	2.0882	2.9608	2.4379	2.4639	0.0183
OVMD-DBiLSTM	2.9563	4.0468	3.4242	3.5038	0.0252
EEMD-DBiLSTM	3.8440	4.9776	4.4507	4.5906	0.0313
OVMD-BPNN	9.6446	10.8783	11.4173	12.2052	0.0712
EEMD-BPNN	10.8439	13.9776	15.1719	14.5906	0.0913
att-SSF-DBiLSTM	3.0428	4.1259	3.0770	3.1289	0.0231
SSF-DBiLSTM	4.3151	5.0807	4.5726	4.1902	0.0293
DBiLSTM	4.8934	5.9861	5.5083	5.5289	0.0371
LSTM	9.5209	10.9208	8.0869	9.6371	0.0603
BPNN	13.3172	14.5634	16.5833	17.0488	0.1052
SVM	15.2071	16.6404	18.1508	20.1216	0.1132

The DM statistics are employed to further test the difference in the forecasting performance between the proposed OVMD-att-SSF-DBiLSTM model and the benchmark models from a statistical perspective. Table 6 shows the values of DM statistics and the corresponding p-values of all models. The p-values are far less than 0.001, then the original hypothesis is rejected on the basis of the DM test. Therefore, there is a significant difference between the proposed OVMD-att-SSF-DBiLSTM model and the other 11 benchmark models, further demonstrating the superiority and robustness of forecasting effectiveness of the proposed model.

In order to further validate the significant results of the model, the nonparametric Wilcoxon signed-rank test is employed on two absolute errors by the compared models. The corresponding statistical test results of the Brent and WTI futures series are presented in Table 7. The p-values of the Wilcoxon signed-rank test for all 11 benchmark models are near zero, falling below the significance level of 0.05. The calculation by the hypothesis test yields H values of 1, indicating that the test rejects the null hypothesis. Consequently, the prediction error of the proposed model is significantly different from the prediction error of the other 11 forecasting models. Besides, in Tables 4 and 5, the error evaluations of the MAE, RMSE, MAPE, SMAPE and TIC by the proposed model are all smaller than those by the other 11 compared forecasting models. Combined with the results of the error evaluation criteria, it can be concluded that the prediction effect of the proposed model is more accurate in forecasting crude oil futures prices.

Additionally, the predictive results of different models are analyzed by the above mentioned MGCS method. According to the MGCS theory, smaller MGCS values are associated with more desirable forecasting results, indicating that the fluctuation behaviors of the prediction are almost consistent with the actual data. Fig. 12 shows the MGCS results between the actual futures price series and the corresponding predictions from the above mentioned 11 comparison models at different q orders and τ scales. In this study, the parameter τ ranges from 1 to 20



(a) Brent (b) WTI

(c) Brent (d) WTI

(e) Brent (f) WTI

Fig. 6. Comparisons of the predicted data and the actual data of Brent and WTI for the forecasting models.

and $q = 2, 3, 4$. When q is fixed, the values of MGCS decrease as the time scale τ increases, indicating that the complexity of information in a series reduces as the length of coarse-grained series steadily decreases. In detail, when $\tau = 1$, the MGCS value is the error value between the original prediction results of each forecasting model and the actual values. As τ increases, the prediction results and the actual values would be coarse-grained, and the corresponding MGCS value reflects the error value of the coarse-grained predictive results and actual values. The findings at different scales can demonstrate the forecasting accuracy of local trends over an extended period for the crude oil futures series.

Tables 8 and 9 depict the specific MGCS results between the actual values and forecasting results from the mentioned benchmark models at the scale $\tau = 1, 5, 10, 20$ for the Brent and WTI futures sequences. The empirical results obtained from the experimental data clearly indicate that the proposed OVMD-att-SSF-DBiLSTM hybrid model outperforms the other 11 benchmark forecasting models at all scales. When $\tau = 1$, the MGCS results of both the Brent and WTI futures sequences are consistent with the conclusions of the other error indicators. However, the MGCS values of the DBiLSTM model exceed those of the SSF-DBiLSTM and EEMD-DBiLSTM models when $q = 2$, $\tau = 5$ and 10 for the Brent futures series. Similarly, in the MGCS numerical results of the WTI futures series, the value of the DBiLSTM model outperforms that

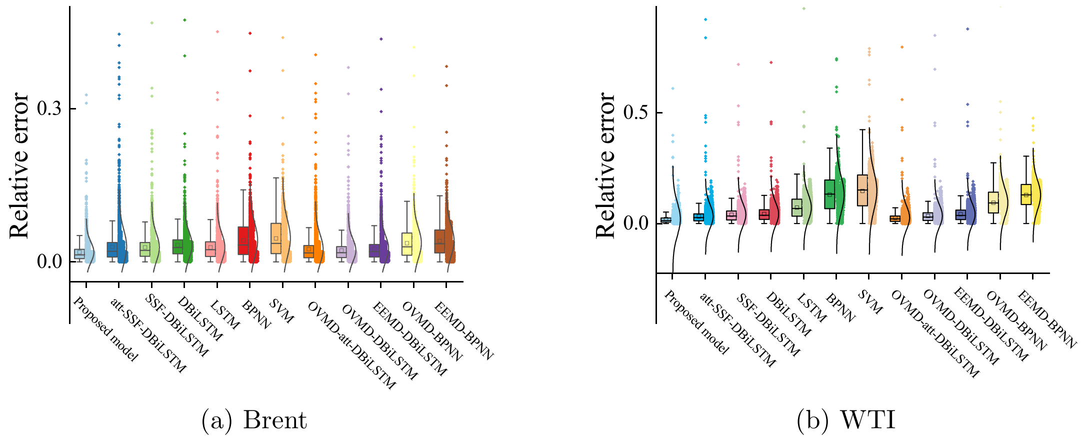


Fig. 7. Forecasting comparison of the evaluation errors from the compared models.

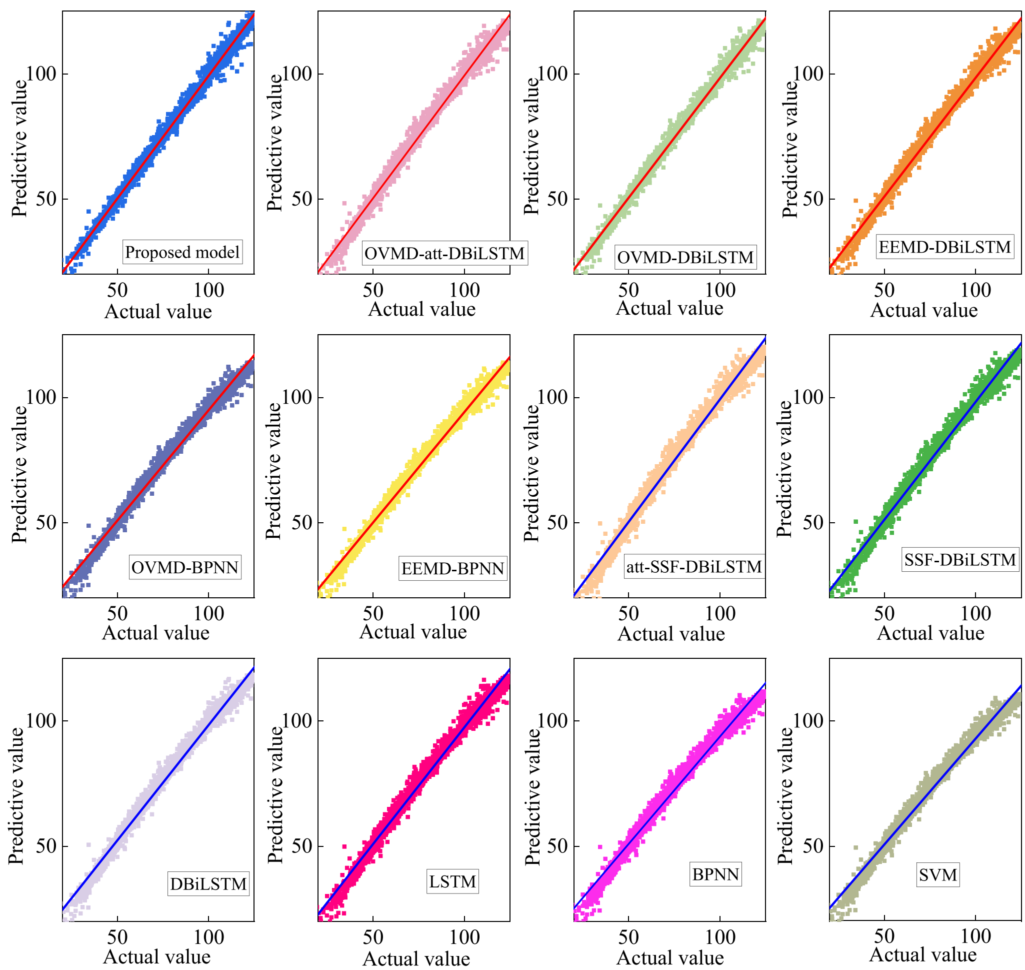


Fig. 8. Fitting results of different models for Brent futures series respectively.

of the EEMD-DBiLSTM model when $q = 4$ and $\tau = 5$. Since the SSF error correction method and the EEMD method primarily aim to enhance the accuracy of fluctuation prediction, but sharp fluctuation points of the original predicted results will be smoothed during the coarse-grained process, and the information and randomness in the original predicted series are decreased and mixed, which may have limitations in the prediction accuracy of local trend. Combined the above results with the MGCS results of $\tau = 1$, it is shown that the prediction accuracy of SSF-DBiLSTM and EEMD-DBiLSTM models have enhanced compared to the results of the DBiLSTM model, but the advantage is not obvious when the scale parameter is expanded. The above results reflect the difference between MGCS method and the other traditional error indicators, which can evaluate the final prediction results from multiple perspectives. Whatever of the scale value is, it does not influence the proposed model has the best predicted performance. In conclusion, regardless of the scale τ value, the predictive accuracy of the proposed model is superior to that of the benchmark models. Then the forecasting capability of the proposed model is further exhibited by the innovative

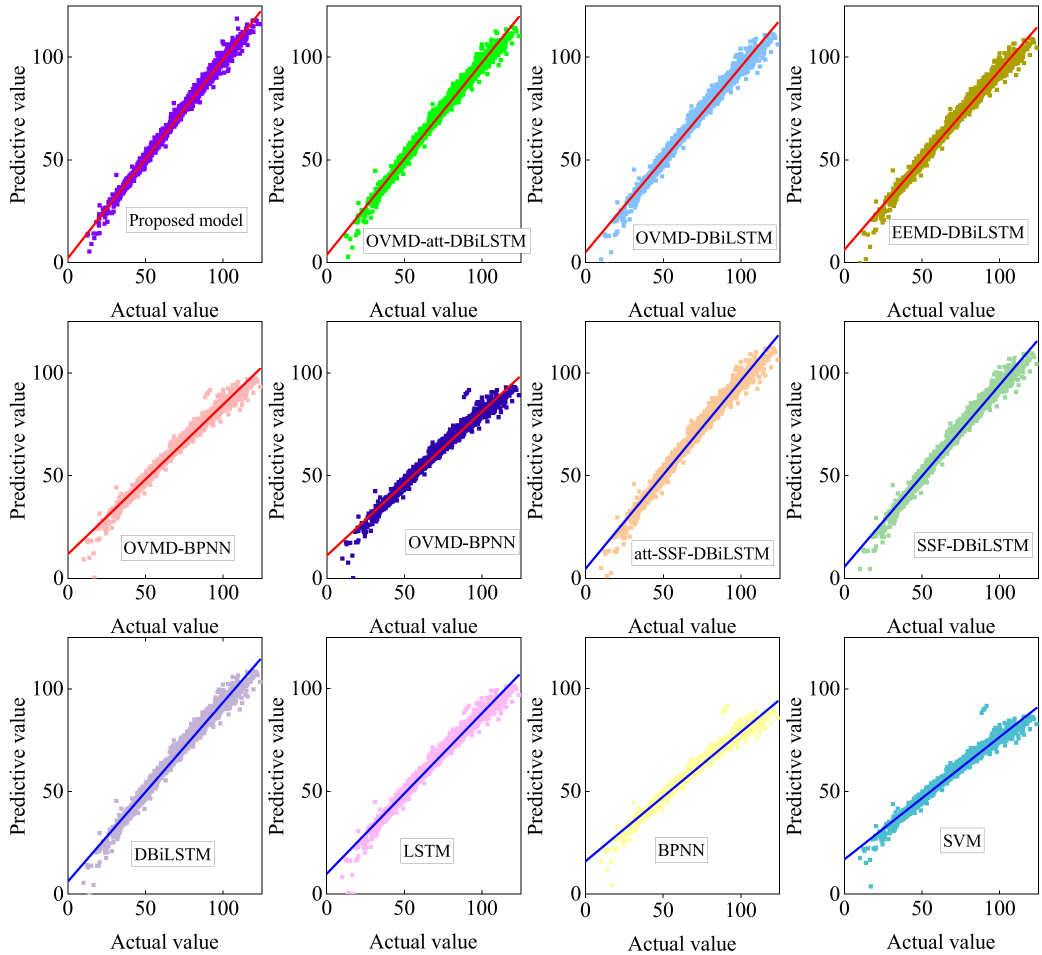


Fig. 9. Fitting results of different models for WTI futures series respectively.

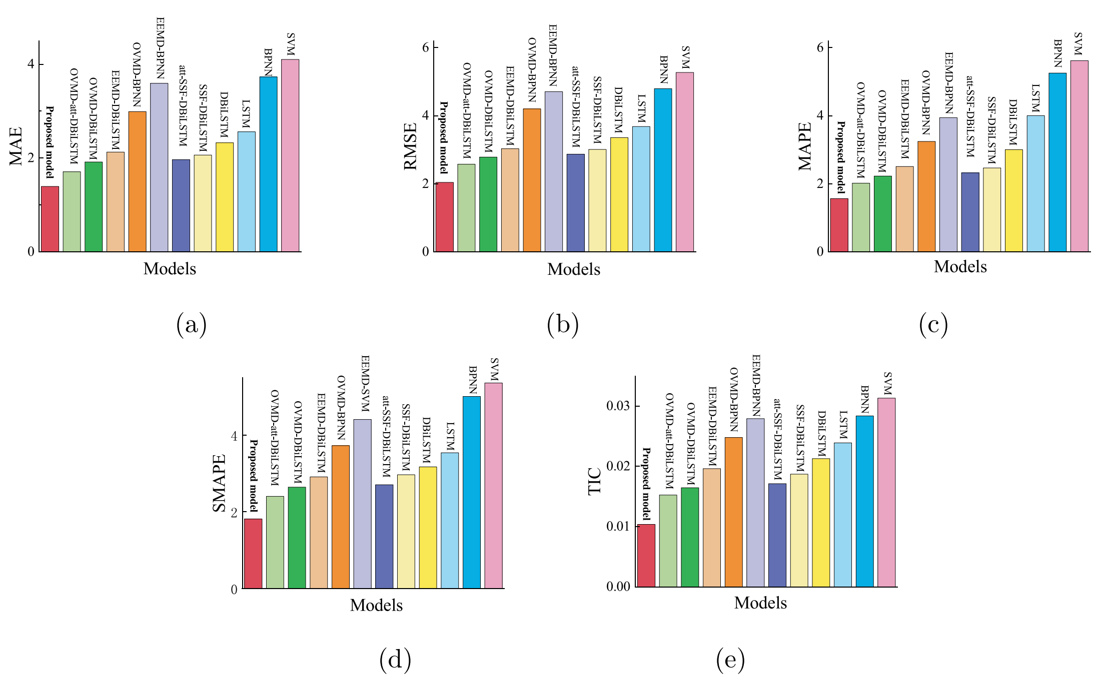


Fig. 10. The evaluation results of the actual Brent futures data points and the forecasting results from different forecasting models.

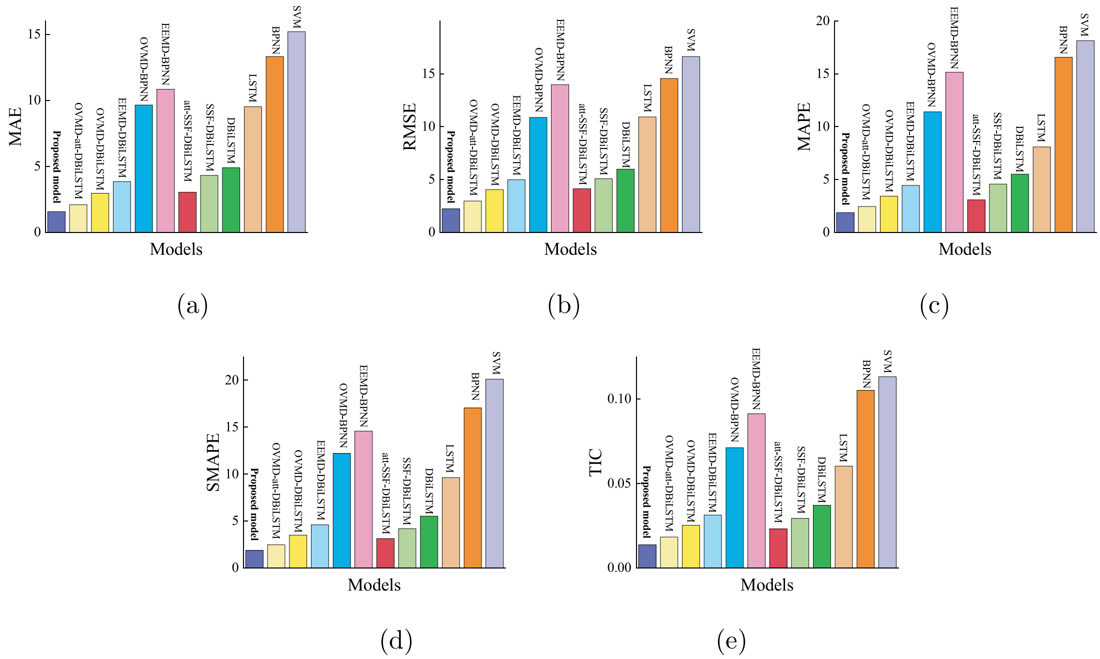


Fig. 11. The evaluation results of the actual WTI futures data points and the forecasting results from different forecasting models.

Table 6
The statistical results of DM test.

Comparison models	Brent		WTI	
	DM statistics	*p*-value	DM statistics	*p*-value
OVMD-att-DBiLSTM	9.5619	0.0005	19.1369	0.0000
OVMD-DBiLSTM	15.9135	0.0000	27.0995	0.0000
EEMD-DBiLSTM	22.0231	0.0000	35.1154	0.0000
OVMD-BPNN	39.3374	0.0000	48.1175	0.0000
EEMD-BPNN	47.2508	0.0000	59.0597	0.0000
att-SSF-DBiLSTM	7.8003	0.0009	28.5124	0.0000
SSF-DBiLSTM	8.0030	0.0001	32.0106	0.0000
DBiLSTM	25.7591	0.0000	41.3045	0.0000
LSTM	34.9433	0.0000	43.9871	0.0000
BPNN	40.3501	0.0000	65.1354	0.0000
SVM	43.4824	0.0000	68.9955	0.0000

Table 7
The statistical results of Wilcoxon signed rank test.

Model	Brent			WTI		
	H	*z*-value	prob.*p*	*H*	*z*-value	prob.*p*
OVMD-att-DBiLSTM	1	4.0231	$5.7446e^{-5}$	1	3.6704	$4.3992e^{-4}$
OVMD-DBiLSTM	1	−12.0169	$4.0347e^{-23}$	1	−16.8029	$3.9740e^{-36}$
EEMD-DBiLSTM	1	−15.1914	$9.0443e^{-30}$	1	−17.8385	$1.9023e^{-38}$
OVMD-BPNN	1	−30.6036	$2.2307e^{-52}$	1	−48.7892	0
EEMD-BPNN	1	−48.3796	0	1	−55.9130	0
att-SSF-DBiLSTM	1	4.0150	$5.9457e^{-5}$	1	11.6704	$6.8385e^{-20}$
SSF-DBiLSTM	1	14.3523	$1.0306e^{-26}$	1	15.9710	$4.7373e^{-33}$
DBiLSTM	1	13.8912	$1.1368e^{-24}$	1	25.3551	$7.9394e^{-43}$
LSTM	1	−16.2128	$2.9061e^{-35}$	1	−28.5392	$1.7182e^{-49}$
BPNN	1	−34.6046	$2.1553e^{-62}$	1	−38.8159	$2.3907e^{-69}$
SVM	1	−52.5182	0	1	−49.5392	0

MGCS method, and the efficiency of the combination of OVMD decomposition method, the error correction algorithm based on SSF and the attention mechanism layer in the proposed model is proved again. Based on the empirical analysis above, the novel developed hybrid forecasting technique significantly enhances the predictive precision of crude oil futures series.

Table 8
MGCS value between the actual data and the corresponding predictions for Brent.

Index	$\tau = 1$	$\tau = 5$	$\tau = 10$	$\tau = 20$
$q = 2$				
Proposed model	95.2532	25.5944	13.1665	7.7832
OVMD-att-DBiLSTM	167.3969	40.0539	22.4877	14.9328
OVMD-DBiLSTM	194.2788	44.9803	26.1186	17.1935
EEMD-DBiLSTM	238.0316	56.9118	33.7119	22.4637
OVMD-BPNN	275.3984	67.7764	37.4033	22.8362
EEMD-BPNN	296.6617	77.8813	47.2212	30.8020
att-SSF-DBiLSTM	226.8981	51.6956	29.3959	18.7340
SSF-DBiLSTM	257.3559	59.7269	33.9399	22.4783
DBiLSTM	263.6554	56.0900	32.6419	22.8664
LSTM	272.6468	61.7348	34.6385	19.4015
BPNN	319.0393	68.9107	37.4370	22.6575
SVM	318.8860	71.9415	38.9921	25.8170
$q = 3$				
Proposed model	25.5258	8.8116	5.2830	3.3365
OVMD-att-DBiLSTM	46.1274	15.0936	10.0127	7.3894
OVMD-DBiLSTM	53.8292	15.5402	9.7475	6.9646
EEMD-DBiLSTM	66.8857	18.9407	12.0526	9.1306
OVMD-BPNN	87.0672	24.5680	15.6141	11.1563
EEMD-BPNN	89.4945	27.4893	17.5113	13.5818
att-SSF-DBiLSTM	63.5138	18.3814	12.0838	9.0969
SSF-DBiLSTM	71.5437	21.1048	13.9212	10.5798
DBiLSTM	73.1860	23.5229	15.5618	12.0534
LSTM	77.0488	21.4382	14.6879	10.6891
BPNN	104.8002	28.7631	16.7183	13.3471
SVM	112.4809	31.1601	19.7776	16.0366
$q = 4$				
Proposed model	16.5125	6.3010	3.2886	2.4019
OVMD-att-DBiLSTM	28.3340	10.1213	6.3154	4.8121
OVMD-DBiLSTM	37.0746	10.5590	6.5361	4.7283
EEMD-DBiLSTM	41.8015	12.5385	8.0396	6.3099
OVMD-BPNN	58.3701	18.9863	13.9893	10.5762
EEMD-BPNN	59.6675	19.2367	14.5648	11.5208
att-SSF-DBiLSTM	38.3176	11.8865	7.8210	5.5599
SSF-DBiLSTM	40.3699	13.3060	8.8930	6.4567
DBiLSTM	45.1686	13.2788	9.8497	7.6383
LSTM	49.3437	14.7169	10.1102	7.8054
BPNN	74.9015	22.3938	16.7811	12.5820
SVM	78.2468	23.9833	17.9161	13.8046

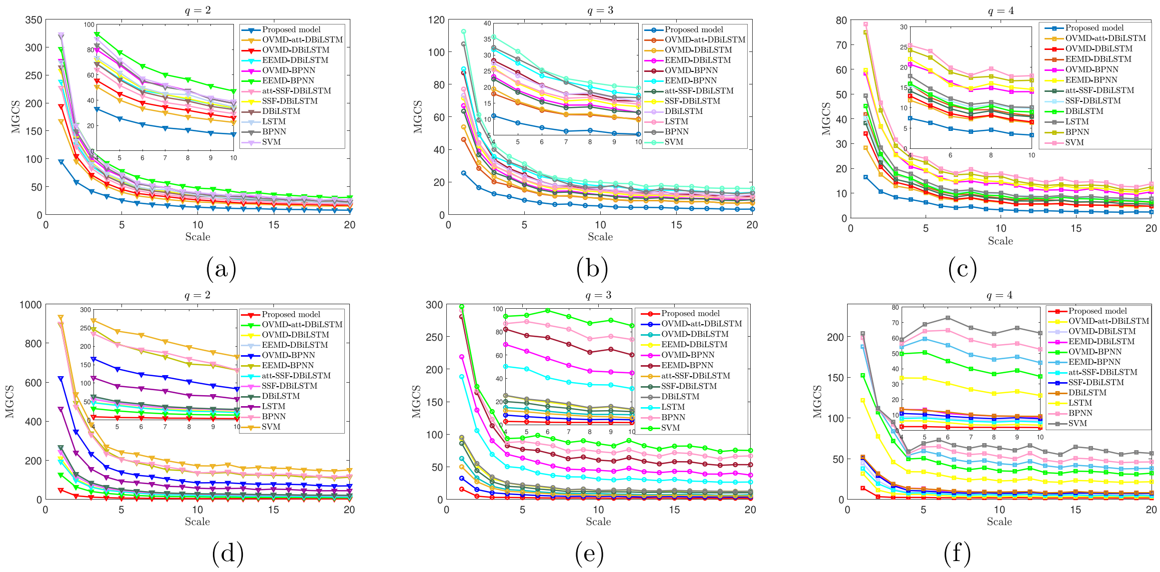


Fig. 12. The MGCS curves of the actual futures data points and the forecasting results from different forecasting models.

Table 9
MGCS value between the actual data and the corresponding predictions for WTI.

Index	$\tau = 1$	$\tau = 5$	$\tau = 10$	$\tau = 20$
$q = 2$				
Proposed model	48.7967	6.6093	3.6479	2.5130
OVMD-att-DBiLSTM	126.7580	24.5003	13.1463	8.5585
OVMD-DBiLSTM	209.7869	40.6352	22.1554	14.2097
EEMD-DBiLSTM	259.5399	47.8011	27.3581	20.6302
OVMD-BPNN	620.9053	138.7233	84.5269	69.0911
EEMD-BPNN	893.1274	206.1474	134.9252	113.5899
att-SSF-DBiLSTM	193.1224	39.1225	19.8233	12.7128
SSF-DBiLSTM	242.6906	45.1147	26.7061	18.3771
DBiLSTM	267.7528	49.7213	28.5134	20.5828
LSTM	464.1216	92.2130	56.9766	44.2932
BPNN	899.2920	204.5644	135.5447	118.5156
SVM	937.7326	241.1729	171.8190	150.5438
$q = 3$				
Proposed model	15.7368	2.7155	2.0426	1.5510
OVMD-att-DBiLSTM	32.4445	7.1238	4.4891	3.8023
OVMD-DBiLSTM	62.8318	13.7487	8.8272	7.1263
EEMD-DBiLSTM	92.9959	21.6924	13.1429	12.5456
OVMD-BPNN	219.0650	63.4386	44.6987	37.3083
EEMD-BPNN	280.6727	76.9133	60.2731	53.3259
att-SSF-DBiLSTM	50.0622	11.2549	6.0309	5.2822
SSF-DBiLSTM	86.0147	18.7339	11.4758	10.3774
DBiLSTM	95.2827	22.2582	13.6067	12.6634
LSTM	188.5251	48.1622	31.3013	26.6682
BPNN	290.2445	88.8635	73.4726	66.8149
SVM	296.1676	94.3459	85.3691	75.1378
$q = 4$				
Proposed model	13.9832	2.3936	1.8104	1.6889
OVMD-att-DBiLSTM	31.6526	6.1228	3.3876	3.1790
OVMD-DBiLSTM	44.4336	9.0148	6.2598	5.0682
EEMD-DBiLSTM	51.9570	13.3604	8.4134	8.1800
OVMD-BPNN	152.4684	50.5613	35.0076	32.1854
EEMD-BPNN	187.7126	59.3774	43.9207	38.1140
att-SSF-DBiLSTM	37.8995	8.0419	5.3705	4.2587
SSF-DBiLSTM	50.8066	10.3852	7.4865	7.2959
DBiLSTM	52.1914	12.8779	9.0590	8.2162
LSTM	121.8045	33.8679	22.6341	21.6707
BPNN	198.5387	64.3867	52.6569	46.0212
SVM	203.9768	68.8336	63.1574	56.5747

5. Conclusion

Forecasting crude oil futures prices becomes crucial since there is a strong connection between severe fluctuations in crude oil prices and shifts in global economic conditions. In this paper, a novel hybrid forecasting model named OVMD-att-SSF-DBiLSTM is built to improve the Brent and WTI futures price forecasting accuracies. In the proposed hybrid forecasting model, the OVMD method is optimized to decompose the original sequences, the SSF algorithm is developed error correction efficiency during the model training, and an attention mechanism layer is constructed as the final layer of the forecasting model to guide the output of predictive results. The effectiveness of the model is illustrated by comparison with five hybrid models including OVMD-att-DBiLSTM, OVMD-DBiLSTM, EEMD-DBiLSTM, OVMD-BPNN, and EEMD-BPNN, and six single models, including att-SSF-DBiLSTM SSF-DBiLSTM, DBiLSTM, LSTM, BPNN and SVM. Furthermore, utilizing linear regression analysis, error evaluation criteria, the MGCS technique, and the DM statistical test, the predictive superiority of the proposed model is illustrated in comparison with the other 11 benchmark models for the Brent and WTI futures prices.

(1) Linear regression and relative error analysis are employed to verify the prediction effect and accuracy of the proposed OVMD-att-SSF-DBiLSTM model. The summary of relative errors in Fig. 9 shows that the proposed OVMD-att-SSF-DBiLSTM model achieves better prediction accuracy than the other forecasting models. In Table 3, compared with those of the benchmark models, the values of a and R^2 with the proposed model are closest to 1, indicating that the proposed model has an excellent prediction effect.

(2) Error evaluation criteria (MAE, RMSE, MAPE, SMAPE, and TIC) and statistical test are employed to demonstrate the prediction performance. In Tables 4–6, and Fig. 11 and Fig. 12 the values of the error evaluation criteria calculated from the OVMD-att-SSF-DBiLSTM model, are the smallest compared with those of the other models. The statistical results of DM test and the Wilcoxon signed-rank test reveal that there are significant differences between the proposed model and the comparative models. Therefore, the proposed model has the best prediction effect for the crude oil futures sequences among all the selected forecasting models.

(3) A new evaluation method, named MGCS analysis, is proposed to assess these models on the synchronization degree for the predicted and actual values. The results show that the MGCS value decreases as values of q and τ increases, and the OVMD-att-SSF-DBiLSTM model yields the smallest MGCS values and the highest prediction accuracy at different q and τ values. This finding further confirms that the proposed model is superior to the other compared models.

(4) From the results of the above mentioned comparison techniques, the OVMD method can effectively help improve accuracy while comparing the pairs of models, such as OVMD-att-SSF-DBiLSTM and att-SSF-DBiLSTM, OVMD-DBiLSTM and EEMD-DBiLSTM. By comparing the forecasting results of relative pairs of models, it is observed that the forecasting model employing the error correction algorithm based on the SSF has the ability to significantly enhance the prediction accuracy of deep learning models. By comparing the forecasting results of the proposed model with those of all the other comparative models, it is demonstrated that utilizing OVMD, the SSF algorithm, and an attention mechanism for model construction can enhance the prediction precision of crude oil futures prices.

The proposed model can precisely predict crude oil futures series and helpful to take actions in advance to reduce the occurrence of financial risk for governments and regulators. In the future, we will continue to improve the decomposition technique of the original series and strive to enhance the prediction accuracy for the energy futures sequences during extreme fluctuation periods.

CRediT authorship contribution statement

Jie Wang: Conceptualization, Methodology, Investigation, Funding acquisition, Writing – original draft, Writing – review & editing. **Ying Zhang:** Data curation, Validation, Software, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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